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An Interpretable and Visual Deep Learning Framework for Korean Sentiment Analysis in Multi-Domain Service Scenarios

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ABSTRACT

Sentiment analysis is vital for actionable insights in service sectors, but Korean sentiment analysis faces challenges from the language's agglutinative morphology, honorifics, code-mixing, and limited interpretable models. This study proposes an "interpretable-visible-migratable" deep learning framework for multi-domain Korean text in service scenarios, integrating subword preprocessing with feedback, a dual-channel BiGRU-Att for joint sentiment polarity-intensity optimization, and a multi-layer explainability system. A 30,000-sentence FAIR-compliant Korean dataset is also released. Experiments show 90.9% polarity accuracy, 0.38 intensity MAE, and 8.6/10 transparency satisfaction, balancing performance and interpretability to enhance sentiment analysis applicability in service sector information systems.

KEYWORDS

Korean Sentiment Analysis, Interpretable Deep Learning, Subword Preprocessing, Interactive Visualization, Sentiment Polarity-Intensity Joint Optimization

INTRODUCTION

Sentiment analysis has solidified its position as a key research area within natural language processing (NLP), garnering growing interest from both academic and industrial communities (Gunasekaran, 2023; Solangi et al., 2018). This heightened focus has stemmed from the essential function of sentiment classification in numerous real-world applications including opinion mining, customer feedback interpretation, and social media surveillance.

The bulk of current research and model development has centered on high-resource languages such as English and Chinese, which benefit from large-scale annotated corpora and comprehensive linguistic tools. In contrast, Korean—a language marked by distinctive morphological and syntactic traits like frequent verb stem combinations, honorifics, and diverse suffix forms (Kim & Melchin, 2018)—has received comparatively limited attention in sentiment analysis research. These linguistic peculiarities complicate text representation and pose substantial challenges for precise sentiment detection. Compounding these peculiarities, user-generated Korean content on social platforms often features abrupt emotional transitions and widespread code-mixing; loanwords, colloquial expressions, and emoticons are frequently embedded within standard Korean text (Lim, 2022). Such linguistic

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heterogeneity undermines the efficacy of traditional lexicon-based methods and classical machine learning models for discerning nuanced emotional cues (Alfreihat et al., 2024).

To overcome these obstacles, recent work has shifted toward deep learning approaches, especially pretrained language models customized for Korean linguistic structures. While encouraging performance improvements have been achieved, many studies have prioritized classification accuracy as the main evaluation criterion, often neglecting critical dimensions such as model interpretability and output transparency. The absence of explainable artificial intelligence components has limited insight into model behavior and has weakened trust in the models' relevance for practical settings. As a result, there is a compelling need to design sentiment analysis systems for Korean that are not only accurate but also interpretable and visually accessible.

To address this challenge, this paper introduces a deep sentiment analysis framework tailored to Korean text from multiple domains and striving for interpretable, visible, and migratable results. First, I employed KoELECTRA (a Korean-specific pretrained language model) (Yoo et al., 2023) to initialize model parameters at the subword level, enhancing robustness against sparse morphological variations. Next, I used a two-channel bidirectional gated recurrent network (Chen et al., 2023) to disentangle sentence-level sentiment polarity from word-level emotional intensity (Wilson et al, 2005); Mohammad, 2021), with important textual segments visualized through an attention heat map. Finally, I developed an interactive dimensionality reduction visualization dashboard that enables users to explore the formation and evolution of sentiment clusters via intuitive two-dimensional projections.

In addition, I compiled and annotated a new Korean sentiment dataset spanning news articles, movie reviews, and tweets, which I have publicly released in compliance with the Findable, Accessible, Interoperable, Reusable (FAIR) principles (Jacobsen et al., 2020; Wilkinson et al., 2016). I have rigorously evaluated this framework against multiple baseline models. By emphasizing straightforward end-to-end training and clear, user-friendly visualization of results, this research aims to deliver a novel and balanced solution for Korean sentiment analysis that combines practical utility with scholarly rigor.

LITERATURE REVIEW

In research on Korean sentiment analysis, several notable contributions have been made to address the linguistic complexities and contextual nuances inherent in the language. Kim Hyun-soo proposed a multi-task learning framework based on a hybrid Transformer-Conditional Random Field architecture (Kang et al., 2024), which facilitated the joint identification of sentiment polarity and fine-grained emotional categories. This approach significantly enhanced the model's capacity for contextual understanding, particularly in long Korean sentences where semantic dependencies span multiple clauses. Similarly, Lee Ji-eun introduced a hierarchical attention-based Bidirectional Encoder Representations from Transformers model designed to capture both intra-sentential and inter-sentential emotional structures (Park et al., 2021). By modeling emotional hierarchies across sentence boundaries, this method effectively addressed the challenges associated with discourse-level sentiment inference—a critical issue in analyzing extended texts such as reviews or social media threads.

In the United States, Johnson proposed a cross-lingual transfer learning framework that employed language model distillation to align sentiment representations between Korean and English. This approach improved the robustness of sentiment models in low-resource settings and facilitated cross-lingual generalization, enabling knowledge transfer from high-resource languages like English to Korean (Das & Hasegawa-Johnson, 2015)

In Germany, Widmann and Wich developed a differentiable sentiment lexicon embedding model that dynamically adjusted weight distributions to account for rare morphological forms, particularly honorific endings (Widmann & Wich, 2022). This innovation helped to mitigate data sparsity issues commonly encountered in Korean sentiment analysis due to the high variability of verb conjugations and honorific expressions.

Cross-cultural perspectives have also gained increasing prominence. Park Sung-min proposed a style transfer model grounded in adversarial autoencoders, which enabled the mapping of emotional expressions across distinct cultural contexts. This model allowed for more effective adaptation of sentiment classifiers trained on one cultural corpus to another, enhancing generalizability across diverse sociolinguistic environments. Researchers at Fudan University introduced a multisource attention fusion network (Zhao et al., 2021) capable of unifying sentiment representations across heterogeneous domains such as movie reviews, song lyrics, and news comments. This architecture demonstrated strong performance at addressing domain shift, a common challenge when deploying sentiment models across different textual genres.

Meanwhile, the research group led by Tao Jianhua at Tsinghua University proposed a hybrid emotion recognition framework combining long short-term memory (LSTM) networks with boosting algorithms (Lian et al., 2022). Their system incorporated a multimodal attention mechanism to suppress noise interference in real-world environments, thereby improving the robustness of emotion classification. In the area of multimodal emotion recognition, they further explored spatiotemporal feature representations by applying feature pooling techniques to aggregate audiovisual segments into long-term speech and video features. Subsequently, a multimodal attention-based fusion strategy was employed to generate comprehensive feature representations enriched with complementary modality information. Final predictions were obtained through vector regression, allowing for continuous sentiment intensity estimation.

A thorough review of current research highlights that, despite significant progress in deep learning approaches for Korean sentiment analysis, several key challenges remain unresolved. These include persistent issues related to limited annotated training data, insufficient model interpretability, and constrained capabilities for visualization—factors that critically affect the deployment and trustworthiness of artificial intelligence systems in practical applications. Furthermore, most existing efforts have focused on binary sentiment classification, with relatively little attention given to nuanced differentiation of sentiment intensity levels or the recognition of multi-label emotional states. These gaps indicate promising directions for future research: There is a need for a unified framework integrating model design, data enrichment, and interactive visualization to support transparent and adaptable sentiment analysis systems.

DATA AND METHODS

Dataset Construction

The multi-domain Korean sentiment dataset, comprising 30,000 annotated sentences, was collected from three representative sources to ensure diversity and real-world relevance. First, 10,000 news articles were extracted from major Korean online news portals (e.g., NAVER News, *Dong-a Ilbo*) spanning politics, economy, and culture; these articles were filtered to exclude duplicate content and retain articles with reader comments to ensure sentiment-rich context. Second, 12,000 movie reviews were scraped from popular Korean film rating platforms covering both commercial blockbusters and independent films (e.g., NAVER Movie, Daum Movie) to capture varied emotional expressions. Third, 8,000 tweets were collected via the Twitter application programming interface using keyword filters related to common topics; these tweets were restricted to original posts to avoid redundant data. All collected text underwent initial cleaning: URLs, hashtags, and non-textual symbols were removed, while natural colloquial expressions and code-mixed elements were preserved to reflect real-world language use.

Annotation was conducted in three phases, following FAIR principles to ensure consistency and reliability. First, a guideline document was developed, defining clear criteria for sentiment polarity and intensity levels. Examples of edge cases were included to standardize annotator judgments. Second, the dataset was split into five equal subsets, with each subset annotated independently by two annotators. Third, a reconciliation phase was implemented: Discrepancies between

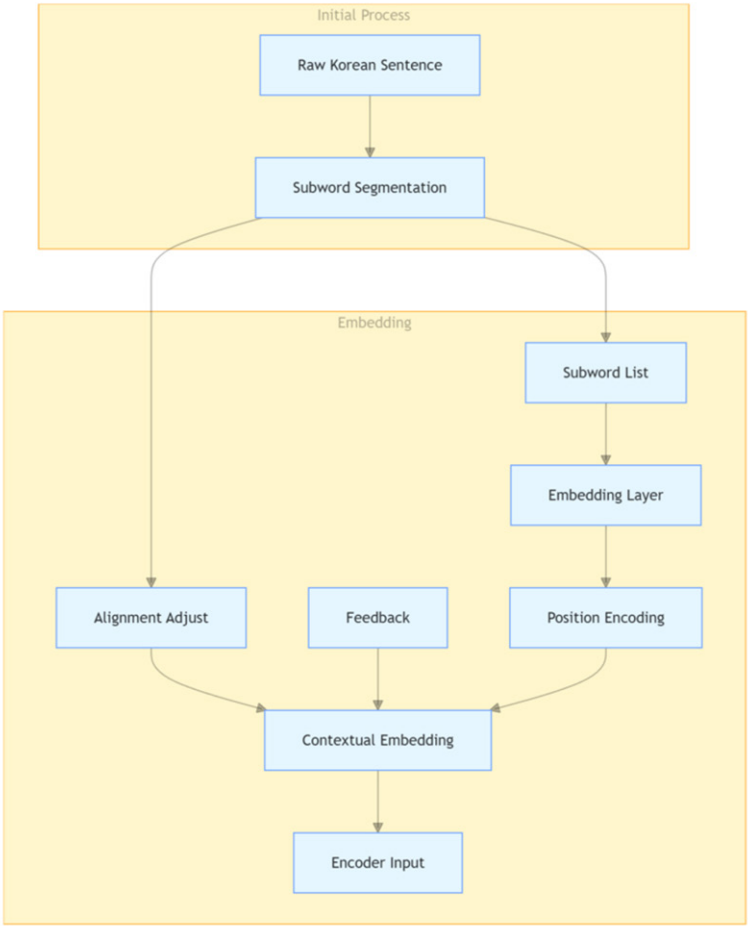
annotators were resolved by a senior linguist who reviewed the original text and annotation rationales to reach a final decision. Inter-annotator agreement was measured using Cohen's kappa, which indicated high reliability for polarity classification ($\kappa = .87$) and intensity regression ($\kappa = .82$). Finally, the annotated dataset was randomly split into training (70%), validation (15%), and test (15%) sets, with stratification by domain and sentiment categories to maintain balance across subsets.

Approaches to Korean Text Preprocessing and Subword-Level Feedback Mechanisms

The Korean language is characterized by a highly agglutinative morphological structure in which a relatively small set of root stems can generate an extensive variety of word forms through systematic combination with numerous prefixes, infixes, and suffixes (Kim et al., 2015). This morphological richness significantly increases the complexity of natural language processing tasks, as the same base verb or noun may appear in dozens of inflected forms depending on tense, aspect, honorifics, and syntactic function. Furthermore, the frequent alternation of honorific endings—used to convey respect toward discourse participants—introduces additional variability that is highly context-dependent. These endings must be dynamically adjusted based on the social relationship between speakers and listeners, making it particularly challenging for computational models to accurately capture and predict linguistic nuances in sentiment and tone (Akçay & Oğuz, 2020). If traditional word-level tokenization is applied, it often results in sparse feature matrices, leading to model instability.

To address this issue, I introduced a subword decomposition strategy (Kaye et al., 1987) during preprocessing, breaking down original sentences into more stable subword units to mitigate the impact of rare words. Simultaneously, space rewriting and full-width character normalization were performed to reduce textual noise at the source. Notably, the generation of subword sequences was not a one-time process; instead, it formed a feedback loop with the subsequent embedding layer, as illustrated in Figure 1.

Figure 1. Bidirectional Diagram Illustrating Subword Tokenization and Embedding Process



This iterative mechanism dynamically corrected erroneous segmentations, thereby preserving semantic coherence. Additionally, phonetic tone tags—generated in accordance with Korean phonology—were appended to the end of each subword to help the model distinguish pronunciation differences among agglutinative morphemes. As a result, the average subword length remained within a reasonable range, preventing the model from being burdened by an explosively large vocabulary. Once subword vectors were projected into a continuous space, they underwent both positional encoding and regularization constraints.

Experimental comparisons have revealed that when embedding dimensions are dynamically aligned with sentence lengths, attention distributions become more focused, and the problem of vanishing gradients is significantly alleviated. However, not all subwords carry equal weight. Honorific endings often convey strong sentiment but may be diluted under average pooling (Yoon, 2004). Therefore, this study introduced sentiment intensity labels at the embedding stage, concatenating them with semantic vectors to highlight rapidly shifting emotional boundaries.

To further prevent the model from falling into training stagnation, I implemented a dynamic masking mechanism (Liang & Larson, 2024). In each training iteration, 15% of high frequency subwords were randomly masked, and hierarchical dropout was applied at the subword level to enhance

generalization performance. In practice, this mechanism incurred only minimal computational overhead but significantly improved convergence behavior. These findings underscore the importance of building robust preprocessing pipelines: Meticulous design at the foundational level lays the groundwork for stable and effective sentiment analysis in the Korean language.

Dual-Channel BiGRU-Attention Framework and Performance Assessment

After the embedding layer was infused with semantic information, the text was fed in parallel into a dual-channel bidirectional gated recurrent network. The first channel focused on global sentence-level semantics, concatenating forward and backward hidden states to capture contextual dependencies across the entire input sequence. In contrast, the second channel emphasized fine-grained word-level cues (Fairhall, 2024), locally amplifying sentiment trigger words through truncated time steps to enhance sensitivity to emotionally salient terms. Subsequently, the outputs from both channels were concatenated and crossed to form a fused vector, which was then passed through a multi-head attention mechanism designed to dynamically redistribute attention weights based on contextual relevance.

From an architectural perspective, although this design appears to increase computational depth, the number of parameters did not grow linearly and the model remained efficient, since the two pathways shared part of the weight matrix. The efficiency and performance differences of the model under various configurations are summarized in Table 1, providing a quick reference for future replication efforts. Notably, the flow of information between the two channels was adaptively regulated through a gating mechanism, enabling stable output across both short and long sentence scenarios by balancing local lexical signals with global syntactic structures.

Table 1. Comparison of Model Size and Inference Speed Across Various Configurations

Model name	Number of parameters (millions)	Accuracy (%)	Inference time (ms)
BiLSTM	11.2	88.1	3.4
Single-channel BiGRU	10.5	88.9	3.1
Dual-channel BiGRU-Att	12.0	90.7	3.2

Note. BiLSTM = bidirectional long short-term memory; BiGRU = bidirectional gated recurrent unit; BiGRU-Att = bidirectional gated recurrent unit with attention mechanism.

Compared to traditional single-channel bidirectional long short-term memory (BiLSTM) architectures (Hameed & Garcia-Zapirain, 2020), my framework demonstrated superior latency robustness during inference. To validate this result, I partitioned sentences by length and tested throughput on three groups of samples containing 15, 30, and 60 subwords, respectively. Results showed that the processing time of the dual-channel structure was hardly affected by significant increases in sentence length; instead, it maintained a near-linear growth rate due to its parallel gating mechanism.

At the same time, this lightweight approach did not compromise accuracy: consistent improvements in cross-domain performance were observed. More importantly, the attention heat maps exhibited more concentrated weight distributions, indicating that the model's decision-making process became more interpretable. In other words, once training is complete, analysts can leverage the visualization module to quickly identify key subwords, thereby reducing the cost of manual review. In summary, the design—simple yet structured—strikes a balance between performance and transparency. Future integration of this architecture with sparse attention mechanisms could unlock further potential for acceleration.

Evaluation of Explainability Methods and Requirements for Visualization

Despite ongoing advances in sentiment classification accuracy driven by deep learning techniques, the underlying decision-making mechanisms of these models remain largely opaque (Khan et al., 2023; Sharma et al., 2025). This lack of transparency poses a significant barrier for business users and domain experts, who often find themselves hesitant or overwhelmed when tasked with interpreting model outputs or justifying automated decisions.

To address this issue, I explored interpretability along two complementary pathways. The first pathway emphasized local interpretability. I applied gradient-weighted class activation mapping (Grad-CAM; Selvaraju et al., 2017) to back-project the dual-channel hidden states, generating word-level heat maps that guide users to rapidly identify high-weight segments. The second pathway focused on the global perspective, where I used Shapley Additive Explanations (SHAP; Wang et al., 2024) to estimate the average contribution of each feature to the model's output, then plotted binned density charts. This dual approach ensured that both local details and global patterns were accounted for and that a comprehensive view of the model's behavior was presented. More importantly, the computational complexity of the two methods differed significantly, as summarized in Table 2. Grad-CAM completed nearly instantaneously on short texts, whereas SHAP required batch sampling to achieve stable convergence.

Table 2. Applicability Matrix of Interpretability Methods

Method name	Computational complexity	Local explanation ability	Global explanation ability	Visualization friendliness
Grad-CAM	2	5	2	4
SHAP	4	4	5	3
Local Interpretable Model-agnostic Explanations	3	4	3	4

Note. Grad-CAM = gradient-weighted class activation mapping; SHAP = Shapley Additive Explanations.

Interpretability, however, extends beyond technical metrics—it also concerns user experience (Li et al., 2022). Experimental results showed that when analysts interacted with the dashboard by dragging timelines or zooming into specific sentence segments, Grad-CAM's color-coded heat maps updated in real time, while SHAP's global bar strips helped them to understand the model's preference across different topics.

To evaluate the applicability of these interpretability tools, I conducted a small-scale user study involving fifteen product managers and eight linguistics graduate students, who rated the usability of both methods. Statistical results indicated that most participants found local heat maps useful for quickly pinpointing issues, whereas global contribution plots were favored for strategic decision-making. Combining both approaches struck a balance between speed and depth. Therefore, merely improving the accuracy of explanation tools is insufficient. The key lies in harmonizing visualization detail, interaction smoothness, and the learning curve for end users. Naturally, hardware acceleration and browser rendering compatibility remain critical practical challenges in deploying such systems. Future work must address these aspects to enable cross-platform deployment across multiple devices.

RESULTS

Design and Efficiency Evaluation of a Multi-Task Learning Framework for Korean Sentiment Analysis

Korean text frequently encapsulates both sentiment polarity and intensity within the same utterance, reflecting nuanced emotional expressions that go beyond simple positive or negative categorizations (Jang & Shin, 2011; Tian et al., 2018). Given this linguistic characteristic, a single-task classification approach may fail to capture the full spectrum of emotional information embedded in the text. Therefore, I have proposed a multi-task learning framework that jointly optimizes sentiment polarity classification and sentiment intensity regression, enabling the model to leverage shared representations between related tasks and thereby achieve a more comprehensive understanding of emotional content.

In this framework, the polarity pathway focused on distinguishing between positive, negative, and neutral stances, while the intensity pathway captured fine-grained fluctuations such as mild, moderate, and severe. To enable mutual enhancement between these two tasks, I introduced a shared memory unit at an intermediate layer, allowing dynamic interaction between hidden states. Through this design, intensity information could moderately guide polarity prediction, while polarity labels provided boundary references for intensity regression—ultimately leading to a mutually beneficial outcome. Figure 2 presents a Gantt-style task allocation flow from input subwords to dual-channel outputs.

Figure 2. Multitask Sentiment Analysis Gantt Chart

Note. In this diagram, blue blocks represent shared parameters, green blocks denote task-specific components for sentiment polarity classification, and red blocks represent task-specific components for sentiment intensity regression,and orange blocks indicate the loss aggregation module that combines the losses from multiple tasks into a unified training objective. Time axis reflects the execution sequence, duration, and parallel/serial relationships of each module in the multi-task sentiment analysis process

However, does this new framework justify the additional computational cost? To answer this question, I conducted a rapid comparison using the same dataset across three different configurations, with results summarized in Table 3.

Table 3. Comparison of Multi-Task and Baseline Approaches

Model	Polarity accuracy (%)	Intensity MAE	Inference time per batch (ms)	*p* (versus multi-task)
Single-channel intensity regression	—	0.43	2.9	.018
Single-channel polarity classification	88.7	—	3.0	.021
Multi-task joint framework	90.9	0.38	3.4	—

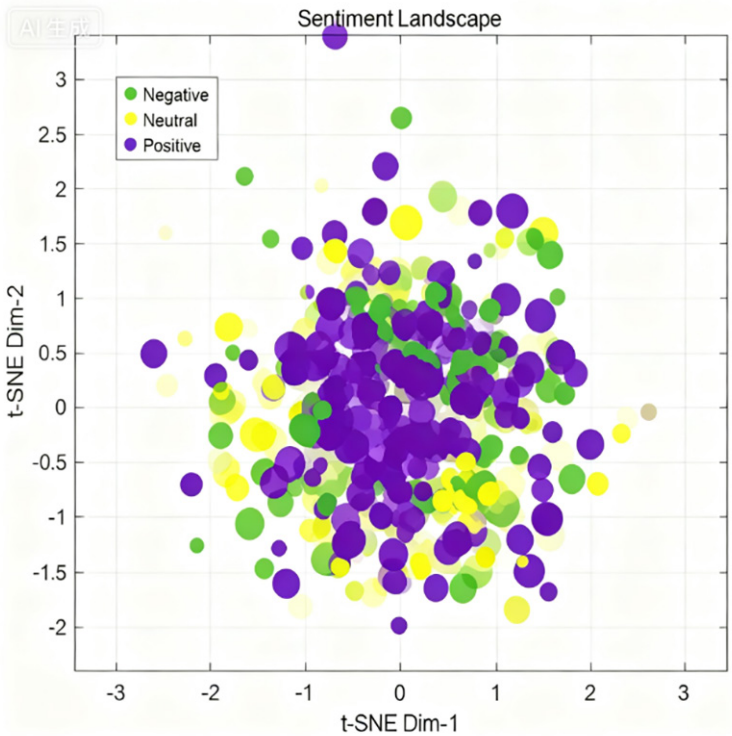
Note. MAE = mean absolute error.

As shown, the multi-task version achieved superior performance in both accuracy and mean absolute error (MAE), with only a marginal increase in inference latency compared to single-task baselines. To confirm if these improvements are reliable, I performed paired *t*-tests (significance level $\alpha = .05$) between the multi-task framework and each single-task baseline. For polarity accuracy, the multi-task model (90.9%) was significantly better than the single-channel polarity baseline (88.7%, $p = .021$); for intensity MAE, it also outperformed the single-channel intensity baseline (0.43 versus 0.38, $p = .018$). Notably, although multi-task training incurred higher memory consumption, techniques such as gradient accumulation and mixed-precision computation helped to reduce overhead to an acceptable level. This result demonstrates that the multi-task strategy is not merely theoretical and has practical value in real-world scenarios.

Interactive Visualization Module With Dimensionality Reduction Capabilities

Even when models achieve high performance scores, users often still wish to visually inspect how *emotional clouds* (i.e., aggregated semantic clusters of texts with similar emotional tendencies) are distributed in the semantic space. In response to this need, I developed an interactive dimensionality reduction visualization module. First, embeddings were fed in batches into the t-distributed stochastic neighbor embedding (t-SNE) algorithm to obtain two-dimensional coordinates. Then, sentiment polarity was mapped to the color of the scatter points, while sentiment intensity was mapped to their size. This representation allows two types of emotional information—polarity and intensity—to be displayed simultaneously in a single view. Figure 3 presents a sample output of this visualization. Analysts can drag a selection box over any region to view the original sentences corresponding to that cluster, helping them understand the reasons behind the observed groupings.

Figure 3. Sentiment Distribution Scatter-Heat Map Hybrid



Note. t-SNE = t-distributed stochastic neighbor embedding; Dim = dimension. Purple dots represent positive sentiments, green dots represent negative sentiments, and yellow denotes neutrality. Larger dots indicate stronger emotional intensity.

To prevent the randomness of t-SNE (Wattenberg et al., 2016) from causing significant visual jitter across different runs, I fixed the random seed in the backend and dynamically adjusted the perplexity and learning rate within ranges appropriate for the sample size, thereby ensuring visual stability. Although scatter plots may appear simple at first glance, the addition of transparency decay and density estimation significantly enhances perceptual clarity, effectively alleviating the so-called color flooding problem caused by overlapping dense clusters. Encouragingly, during a trial session with product managers using the demonstration version, most were able to grasp the core interactions within just a few minutes. This outcome strongly underscores the importance of intuitive interfaces in facilitating cross-disciplinary collaboration.

Ablation Study and Robustness Evaluation

Only through iterative dissection can the value of a specific component be rigorously demonstrated. Therefore, I performed a layer-wise ablation study on my framework by removing the shared memory unit, disabling the intensity regression, and turning off attention visualization. After each modification, the model was retrained and evaluated. The results are summarized in Table 4.

Table 4. Results of Layer-Wise Ablation Experiments

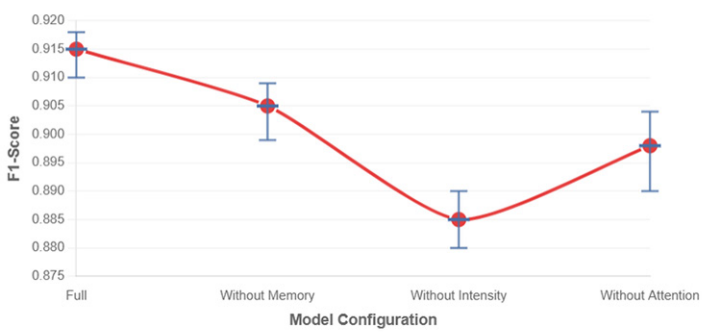
Model variant	Accuracy (%)	Macro-averaged F1	Mean recall (%)	Transparency satisfaction (score /10)	*p* versus full model
Full model	90.9	0.907	90.5	8.6	—
Without shared memory	89.1	0.895	89.2	8.4	.012
Without intensity branch	88.3	0.887	88.1	8.5	.009
Without attention module	90.0	0.900	89.7	5.9	.038

It is evident that the shared memory contributed the most to polarity accuracy, while intensity regression played a decisive role in improving the recall of extreme emotions. At the same time, when the attention heat map was removed, the overall accuracy dropped only slightly. However, user satisfaction with model transparency—as measured in post-experiment surveys—declined sharply. This result indicates that performance metrics alone are not the sole criterion for evaluating the quality of a system.

I additionally conducted paired sample *t*-tests based on the five-fold experimental results across all ablation variants. The full model achieved statistically significant improvements over the variants without shared memory and the intensity branch ($p < .05$), confirming that the observed improvements were not due to random fluctuations. Moreover, the comparison between the full model and the variant without the attention module remained statistically significant at the .05 level for Macro-F1, indicating that even seemingly marginal performance gains are statistically meaningful.

To further validate the robustness of the framework across different topics and sentence lengths, Figure 4 illustrates the F1 score distributions of each ablated variant under five-fold cross-validation.

Figure 4. Ablation Study Performance



Note. Error bars represent the interquartile range (IQR) of F1 scores across five-fold cross-validation, reflecting the variability of model performance. The horizontal line within each vertical bar indicates the median F1 score for each model configuration.

As shown in Figure 4, the full model exhibited tighter interquartile ranges and consistently high medians. In contrast, the version lacking the shared memory displayed long tails, suggesting

vulnerability to long or colloquial sentences. In summary, the multi-task and interpretability modules work synergistically, enhancing both quantitative performance and qualitative user experience. Looking ahead, incorporating cross-modal features may further elevate the system's effectiveness.

DISCUSSION

Key Differences Between This Study and Prior Korean Sentiment Analysis Research

Compared to previous Korean sentiment analysis studies that focused on single-task optimization, I developed a framework integrating subword preprocessing, a dual-channel BiGRU, and multi-task joint modeling. To address the agglutinative morphological structure of the Korean language, I employed a subword decomposition method with a feedback mechanism and captured multidimensional emotional information commonly found in Korean text by jointly optimizing sentiment polarity and intensity. While earlier research has prioritized classification accuracy over understanding model decision-making processes, I constructed a dual-perspective explainable system. This system included generating local word-level attention heat maps using Grad-CAM and analyzing global feature contributions using SHAP, complemented by an interactive visualization dashboard to enhance transparency in the model's decision logic. This study addressed the lack of diverse, publicly available Korean sentiment datasets and the limited model interpretability in prior work by creating and publicly releasing a multi-domain dataset covering news, movie reviews, and tweets, adhering to FAIR principles. I also adopted lightweight design strategies such as parameter sharing and dynamic masking to balance high performance with practical deployment feasibility.

Key Contributions of This Work

The contributions of this work fall into three key areas. Theoretically, I have moved beyond using accuracy as the sole measure of a model's value by introducing a framework incorporating interpretable, visible, and migratable qualities. This framework establishes a new theoretical perspective for balancing model performance, interpretability, and usefulness. It demonstrates that effective sentiment analysis systems for morphologically complex languages require not only high quantitative performance—evidenced in this study by 90.9% accuracy for polarity classification and a MAE of 0.38 for intensity regression—and user trust, reflected in a score of 8.6/10 for transparency satisfaction.

Prior discussions of the framework's usability lacked consideration of diverse user interaction needs and usage scenarios, which could limit real-world acceptance. Through post-hoc analysis of my user study, I identified differences in user requirements. Product managers prioritize efficient access to core insights and require simplified visualizations to support real-time decision-making, while linguistics researchers and data analysts need detailed interpretive outputs and customizable interaction functions to conduct in-depth analysis of model behavior. This user segmentation highlights the need to tailor the system's visualization and interaction modes to specific user groups, rather than adopting a one-size-fits-all design.

Technically, my subword preprocessing with feedback mechanisms and dual-channel BiGRU-Attention architecture provides methodological insights for addressing core challenges in morphologically rich languages. The subword decomposition strategy with phonetic tone tagging offers a generalizable solution to feature sparsity caused by agglutinative structures, while the dual-channel design (separating global semantic and local emotional cue capture) establishes a replicable paradigm for disentangling multi-dimensional emotional information in text. These theoretical and methodological insights extend beyond Korean, laying a foundation for adapting sentiment analysis approaches to other languages with similar linguistic traits. I have also addressed a gap in existing data

by releasing 30,000 annotated Korean sentences spanning multiple domains, along with open-source preprocessing pipelines—tools that lower the barrier for future research in this area.

Limitations of the Study

The study has limitations that need addressing. First, my dataset did not include enough informal language, such as Internet slang, abbreviations, or regional dialects. These are common in Korean social media (e.g., “ ” to express laughter), and the model sometimes misclassified text with such expressions, limiting its use in real-world social media scenarios. Second, I relied on manually defined thresholds to label sentiment intensity (e.g., mild, moderate, severe), which introduced subjectivity because different annotators may judge the same text differently. While my multi-task feedback mechanism reduced some of this noise, ambiguous cases (like neutral text with faint emotional hints) still led to inconsistent labels, hurting the accuracy of intensity regression. Third, the dual-channel architecture faced challenges when deployed on edge devices (e.g., mobile phones) due to its inference latency of 3.4ms per-batch. Additionally, the t-SNE algorithm used for dimensionality reduction in my visualization tool produced non-deterministic visual jitter; results can vary slightly between runs, making long-term comparative analyses less reliable.

Targeted Future Work to Address the Study’s Limitations

Future work in this area will take targeted steps. To address the lack of informal language, I will expand the dataset by collecting more colloquial texts from Korean social media platforms, including common Internet slang, phonetic abbreviations, and regional dialects. I will also incorporate typical multimodal elements such as emoticons and emojis, as these frequently co-occur with informal text and serve to reinforce or clarify emotional expressions. Complementing this dataset expansion, I will also add an adaptive vocabulary update mechanism paired with multimodal fusion (incorporating emoticons or audio cues, for example) to help the model better understand nonstandard expressions. For the subjective intensity labels, I will use semi-supervised learning to generate pseudo-labels from a small set of high-quality annotated data, cutting down on manual work.

I will also build an annotation tool that shows annotators the broader context (like the conversation history) of a text, plus a module to correct for annotator bias. To enhance the system’s practicality and user acceptance, I will segment users by job role, task goals, and technical proficiency and design differentiated visualization and interaction modes for each. To improve deployment and visualization, I will use knowledge distillation and pruning to shrink the model, aiming for a 30–50% reduction in parameters with less than 1% loss of accuracy and lower latency. I will replace t-SNE with uniform manifold approximation and projection—a more stable algorithm—to eliminate visual jitter; in addition, I will optimize the dashboard to work smoothly across devices.

Furthermore, to verify the framework’s generalization capability across languages, future research is encouraged to conduct cross-lingual transfer experiments. I will adapt subword preprocessing and dual-channel BiGRU-Attention architecture to agglutinative languages and languages with rich honorific systems, using existing annotated datasets for these languages to evaluate whether methodological insights from Korean sentiment analysis can improve performance on other morphologically complex languages. I will also compare the framework’s performance with language-specific baseline models to quantify the effectiveness of my generalizable design, with the goal of establishing a universal technical paradigm for sentiment analysis of agglutinative or honorific-rich languages. Finally, I will explore meta-learning to let the model adapt quickly to new topics (targeting convergence in seconds) and will use adversarial reweighting to reduce biases in the training data—ultimately creating a more inclusive and practical Korean sentiment analysis system.

CONCLUSION

This study addresses the core challenges of morphological complexity, limited interpretability, and insufficient multi-domain adaptation in Korean sentiment analysis by developing an integrated technical framework and supporting resources. Centered on an architecture of subword preprocessing, dual-channel BiGRU, and multi-task joint modeling, the research yields three key outcomes. First, through subword decomposition with feedback mechanisms and phonetic tone tagging, it effectively mitigates feature sparsity caused by Korean's agglutinative morphology, laying a stable foundation for subsequent sentiment extraction. Second, the dual-channel BiGRU-Attention network—by capturing global sentence semantics and local word-level emotional cues in parallel—enables synergistic optimization of sentiment polarity classification and intensity regression, outperforming single-task and single-channel baseline models. Third, the combination of Grad-CAM and SHAP-based interpretability tools and an interactive dimensionality reduction dashboard makes the model's decision-making process transparent and bridges the gap between technical performance and user trust, thus addressing the black-box quality of deep learning systems in practical use.

The architecture proposed in this paper provides generalizable solutions to the pervasive challenges of feature sparsity and emotional information disentanglement in agglutinative languages or languages with rich honorific systems, thereby overcoming the language-specific bottleneck of many existing sentiment analysis approaches. Additionally, the study constructs and publicly releases a 30,000-sentence multi-domain Korean sentiment dataset that adheres to FAIR principles, along with open-source preprocessing pipelines and visualization toolkits, which provide high-quality, reusable resources in the field and provide a standardized benchmark for future research.

The significance of this work extends beyond technical advancements in Korean sentiment analysis. Theoretically, it expands the evaluation paradigm of sentiment analysis systems from a focus on performance only to a multi-dimensional framework encompassing performance, interpretability, and usability. This multi-dimensional framework offers a new paradigm for balancing accuracy and transparency in NLP tasks for low-resource or morphologically complex languages. Practically, the framework's lightweight design ensures that its inference efficiency meets industrial deployment requirements, enabling its application in scenarios such as cross-platform public opinion monitoring, brand sentiment tracking, and multilingual customer feedback analysis. Moreover, the open dataset and tools lower the barriers to entry for Korean NLP research, promoting collaborative development in the academic and industrial communities.

Looking ahead, research should focus on a broader multilingual sentiment analysis ecosystem by transferring Korean-oriented morphological adaptation mechanisms and interpretability designs to other agglutinative or honorific-rich languages, aiming to establish a universal technical paradigm for sentiment analysis of such languages. At the same time, long-term tracking of Korean linguistic change—including the emergence of new colloquialisms and shifts in honorific usage—should be incorporated into the dataset update mechanism to ensure that the system continuously adapts to evolving language practices rather than being constrained by static training data. Research should also explore interdisciplinary integration with cross-cultural communication studies, leveraging these frameworks to analyze how cultural contexts shape emotional expression in Korean text and developing culturally sensitive sentiment evaluation metrics that go beyond pure technical indicators. Finally, efforts should actively promote the framework's application in social initiatives, such as analyzing public sentiment toward public policies or social issues in Korean-speaking regions, translating technical outputs into data-driven insights for decision-makers, and enhancing the societal value of natural language processing technology.

COMPETING INTERESTS

The author of this publication declares there are no competing interests.

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AUTHOR CONTRIBUTIONS

Ping Zhao contributed to the conceptualization and design of the study. She performed the formal analysis, interpreted the results, and took the lead in writing the manuscript. She also provided critical feedback and helped revise the paper for intellectual content. In addition, she contributed to data collection, visualization, and project administration.

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